Enetic
For Energy Projects

MAHMOUD GILANY	PROFESSOR OF ELECTRICAL POWER ENGINEERING. Dept. of Electrical Power Engineering, Faculty of Engineering, Cairo University, Egypt. Energy Consultancy Section Head @ Enetic Company e-mail: prof.gilany@enetic-eg.com
Education	○ Ph.D. Calgary University, Canada (1993) ● Field of Study: A Microprocessor Based Relay for Parallel Transmission Lines. ○ M. Sc. Cairo University, Egypt (1989) ● Field of Study: Digital Relaying of Transmission lines. ○ B. Sc. Cairo University, Egypt (1987) ● B.Sc. Thesis on Motor Speed Control. (Top of the Class).
Academic Summary	○ 2006-Present : Full Professor, Cairo University. ○ 2013-Present : Professor (Part-time) at the American University in Cairo. ○ 2003 – 2013: Assoc. Professor at College of Technological Studies, (CTS Kuwait). ○ 1995 – 2003.: Assoc. Professor (Part Time) at the American University in Cairo. ○ 1999-2006: Associated Professor, Cairo University ○ 1993-1999 : Assistant Professor, Cairo University ○ 1989 -1993: Graduate Research Assistant and Graduate Teaching Assistant, Calgary University.
Academic Courses Taught	○ **Cairo University** ● Power System Protection ● Digital Protection ● Electrical Power Plants ● Electrical Substations ● Electrical Installations ● Energy Utilization ● Renewable Energy Sources. ○ **American University in Cairo (AUC)** ● ENGR-318 (Electricity for Non Electrical Engineers). For Civil and Mechanical Dept. ● EENG 3601 (Power and Machines) For Communication Dept. ● PENG 218/2411 (Electrical Engineering) For Petroleum Dept. ○ **College Of Technological Studies, Kuwait**

	● Electrical Substations ● Operation and Control of Power Systems ● Programmable Logic Controllers PLC ● Electrical Maintenance ● Electrical Materials ● Electrical Machines ○ **Other Courses** ● Renewable Energy (Open Arab University, Kuwait) ● Statistics (Kuwait University, College for Women)
Work Experiences	Consultant and Supervisor for the Electrical Works in the following Projects: 1. **Chairman of the National Committee responsible for updating Egyptian standard specifications. 2024-now** The committee is responsible for updating Egyptian specifications for the following products: ➢ Medium voltage switchgear ➢ Medium voltage cables ➢ Power transformers ➢ Low voltage cables and wires ➢ Low voltage panels and equipment ➢ Pipes, ducts, and boxes ➢ Lighting units ➢ Electrical distribution circuits and accessories 2. **Member of the Higher Committee for the Accreditation of Professional Engineers at the Engineers Syndicate. 2024-now.** 3. **Establishing a master plan for the development of the Libyan electricity network. 2023 – now** ➢ Establish a master plan for an advanced and economical electrical energy system that operates efficiently at the level of generation, transmission, and distribution to meet current and future demand. ➢ Developing the transmission and distribution systems to improve service and reduce interruptions. ➢ Modernizing the network with advanced monitoring and control centers. 4. **Technical Adviser for the Egyptian Ministry of Electricity for Power Stations in National Projects. 2020- to date** ➢ Impact and Grid connection studies including: ➢ Transients' stability ➢ Critical Fault Clearing Time Calculations ➢ Reliability analysis

- Reactive power capability.

5. **Fault analyzer for different major failures/Fires in different Egyptian substations.**

6. **Technical Consultant for Khalda Petroleum Company (2023-date):**
   Engineering Design and Specification for MV Network. The work includes:
   - State the required technical specs for (OH Conductor, Insulators, Cut-out Fuses, Stay guy, etc
   - AC Load flow calculation
   - The min. and max. short circuit Calculations
   - Earthing system Design
   - Protection System Coordination study
   - Estimating the required number/type of Line Fault Indicator
   - Investigation for Load Loss Events.

7. **Vendor Accreditation activities:**
   - Accreditation of High voltage insulators factory for Khalda Petroleum company
   - Accreditation of a medium voltage Transformer factory in Turky (AL Makson) for TUV

8. **Chronic Care Center, Gedda, Saudi Arabia (100 Beds) -2015**
   - Design the full electrical network
   - Design the light current disciplines for the hospital.

9. **National Cancer Institute, Cairo - 2013-2014**
   The Hospital Has 200 Beds and 13 Operation Rooms.
   - Design the full electrical network (13MW)
   - Design the light current systems
   - Design the emergency systems.

10. **US Army Corps of Engineers, F16 depot facilities, Egypt. 2000-2003**
    Preparing electrical shop drawings
    - Supervise the implementations carried by Encorp International.
    - Relay settings
    - Lighting and power loads,
    - Light current systems,
    - Installations in hazard areas,
    - 11 kV distribution network,

- Earthing system,
- Short Circuit re-calculation,
- Coordination study.

11. Cairo University - 1999-2003

- Member of the "University Consultant Engineering Supreme Committee.
- Preparing and Reviewing the electrical technical specifications Guide for the university projects
- Design and Supervise the electric projects at more than 20 Buildings in the university including Labs, Student Hospital, Student Housing, Studying Halls, Gardens, etc.

12. Exxon-Mobil - 2002

- Design the Electrical Network of the Fire Fighting System in Mostord terminal.
- Design of the control and co-ordination systems between Electric , Jockey and Diesel pumps.
- Inspector of the fire accidents occurred in different stations.

13. Capital Stock Authority - 1998

- Evaluation of the value of Transferring the Technology of Dry Transformers to Egypt

14. AMONSETO Textile Company (10th of Ramadan City) 1997-1998

- Design the medium and low voltage networks
- Design the Tie-connection between the company standby generating station (21 MW) and the Egyptian Grid.

15. Khalda Petroleum Co., West Desert-Egypt - 1997

- Studies for the installation of the fourth Turbine in SALAM base (4.125 MVA), 200 Oil wells.
- Setting of Protective Relays (Digital and Analog)
- Testing, and Coordination of all relays.
- Short Circuit Currents calculation.
- Inspection of 3.3 kV overhead distribution network problems.

16. ESSO Company, Egypt - 1995-2002

- Design and supervise the installation of the electric networks and Pump areas in ESSO Terminals at Mostord & El-Max city.

17. Scientific Group, Egypt - 1995-2000

- Design the electrical (Power and light current) networks of NORAN Resort at the Northern Coast (owned by the scientific group for engineering Projects).

18. Cement Companies, Egypt - 1996

- Studies for a power factor improvement project at The National Cement Company (44 MVA).

Laboratory and Courses Development

- Member Of the Proposal Implementation Committee (Engineering Education Development Project In Egypt)
- Supervising the implementation of three Electrical Labs in Port-Said Faculty of Engineering (Bids Preparation, Equipment Testing, etc.).
- Member of the ABET Accreditation team in two Faculties: AUC (Egypt-2014) and CTS (Kuwait-2002)

Training Courses Presented

(Presented More Than 3000 Training Hours, In 7-Countries)

Course Title	FREQ	DUR	Client/Country	Year
Digital Protection	5	5	Ministry of Electricity/Egypt	1995-2003
Grounding	1	5	Arab Contractors/Egypt	1999
Protective Relays	3	5	Ministry of Electricity/Egypt	1999 2001
Power Electronics	2	5	Energy Research Center/Egypt	2002
Electrical Installation	1	5	KSE/Kuwait	2004
Internal Lighting Design	1	5	KSE/Kuwait	2004
Electrical Safety in Buildings	1	5	KSE/Kuwait	2008
Solar Energy	1	5	KSE/Kuwait	2008
MV Transformers	1	5	KSE/Kuwait	2009
Electrical Installation	1	5	KSE/Kuwait	2009
Managing and execution of Electrical Maintenance	1	5	KSE/Kuwait	2009
Sound Systems in Mosques	2	5	Ministry of Awqaf/Kuwait	2009
Electrical Installation	1	5	KSE/Kuwait	2010
Electrical Safety in Buildings	1	5	KSE/Kuwait	2010

Power system Protection	1	5	KSE/Kuwait	2010
Electrical Safety in Buildings	1	5	KSE/Kuwait	2011
Synchronous Generators	1	5	KSE/Kuwait	2011
Predictive and corrective Maintenance	1	5	KSE/Kuwait	2011
Electrical Safety in Buildings	3	5	KSE/Kuwait	2011
Power system Protection	1	5	KSE/Kuwait	2011
Electrical Maintenance	1	5	KSE/Kuwait	2012
Electrical Equipment Sizing for Expertise	1	5	Ministry of Justice/Kuwait	2012
Protective Relay Maintenance Application & Testing	1	5	KPC/Kuwait	2013
Electrical Installation	1	5	KSE/Kuwait	2012
Innovative Solutions	1	5	KSE/Kuwait	2012
Street Lighting	1	5	KSE/Kuwait	2012
Electrical Safety in Buildings	6	5	KSE/Kuwait	2012
Electrical Maintenance	2	5	Touristic Enterprises/Kuwait	2012
Fire Alarm and Fire Fighting	1	5	KSE/Kuwait	2013
Electrical Installation	4	5	KSE/Kuwait	2013
Electrical Safety in Buildings	3	5	KSE/Kuwait	2013
Smart Homes	1	5	KSE/Kuwait	2013
Power system Protection	1	5	KSE/Kuwait	2013
Innovative Thinking	1	5	KSE/Kuwait	2013
Electrical Wiring, Regulation, Installation and Inspection	1	5	KahraMaa/Qatar	2013
HV & MV Substation Design, Testing and Maintenance	1	5	KahraMaa/Qatar	2013
Power Transformers Diagnostic and Monitoring	1	5	KahraMaa/Qatar	2013
Power System Protection	1	5	KahraMaa/Qatar	2013
Power Cable Standard &Testing	1	5	KahraMaa/Qatar	2014
Electrical Faults, Causes, Detection and Remedies	1	5	KahraMaa/Qatar	2014
Distribution SF6 RMU	1	5	KahraMaa/Qatar	2014
Electrical Drawing, Schematic and Wiring	1	5	KahraMaa/Qatar	2014
Electrical Installation	1	5	KahraMaa/Qatar	2014
Electrical Power Transmission and Distribution	1	5	KahraMaa/Qatar	2015

Electrical Drawings and Schematic	2	5	KahraMaa/Qatar	2015
Electrical Professional Engineer	5	5	KahraMaa/Qatar	2016
Managing & Controlling Power Losses in Electrical Network	2	5	Muscat Distribution/ OMAN	2015
Electrical Substation Design Fundamentals	1	5	KahraMaa/Qatar	2015
Transmission Substation Inspection and Reporting	1	5	KahraMaa/Qatar	2015
Power System Restoration Strategy during Emergency & Black Out	1	5	Different delegates, DUBAI	2015
Control and Reduction of Power Losses in Electrical Networks	1	5	Different delegates, Saudi Arabia/Spain	2015
Load Flow, Short Circuit Calculations & Protection Relay Coordination	1	5	Muscat Distribution/ OMAN	2015
Patrolling, Route Identification and reporting	1	5	KahraMaa/Qatar	2016
MV CB testing and Maintenance	1	5	KahraMaa/Qatar	2016
Planning of Electric Power Distribution	1	10	KahraMaa/Qatar	2016
Fundamental of Power System Equipment	2	5	KahraMaa/Qatar	2016
Electrical Power T&D Principles and Applications	1	5	KahraMaa/Qatar	2016
Power Quality Monitoring, analysis and Mitigation	1	5	KahraMaa/Qatar	2016
Protective Relaying	1	5	Qatar Petroleum/Qatar	2016
Electrical Utility Communications & Smart Grid Technologies	1	5	Different delegates, Saudi Arabia / BAHRIN	2016
Electrical Distribution Network: Engineering, Design & Planning Techniques	1	5	Public Course/Dubai	2016
Electrical Transformers, Malfunctions, Insulation, and Protection	1	5	Different delegates, Saudi Arabia/Istanbool	2017
Power System Analysis	1	5	ADNOC, Abo-Dhbie	2019
Electrical Installation	1	5	KSE- Kuwait (online)	2020

Electrical Drawings	1	5	KSE- Kuwait (online)	2021
Voltage Regulation	1	5	Apex UAE (online)	2021
Power System Protection	1	10	Apex UAE (Dubai)	2021
Power System Protection	1	5	KSE- Kuwait (online)	2022
Electrical Power Engineering	1	10	Saudi K (online)	2022
Substation Earthing Design	1	15	Egypt (Sphinx factory)	2023
Substation Earthing Design	1	20	Bahrain (Electric Distribution Company)	2023

Technical Development Courses	○ Teaching Enhancement (Center for Learning and Teaching, AUC-2015) ○ Teaching Evaluating - 2011 ○ WIDS - 2010 ○ Photoshop - 2006 ○ SPSS - 2005 ○ AutoCAD - 2005 ○ Writing Technical Notes - 2004 ○ Microprocessor Based Relaying – 1992.
Advising	**Supervised 16 Ph.D. theses and 29 Master's theses.** **Ph.D. advisees (with the first employment after graduation**) 1- Mahmod Esam Atta (2022) ● Proposed Protection Schemes for Detecting and Diagnosis Incipient Broken Bars and Bearing Faults in Induction Motors 2- Mahmod Dawood (2021) ● Improving Protective Schemes of Distribution Networks and Microgrids 3- Galal Abd Elaal (2021) ● Enhancement of Electric Vehicles Integration in Residential Distribution Sector 4- Loai Mohamed (2021) ● An accurate technique for supervising distance relays during power swing. 5- Basem Abd-El-Hameed: (2020) ● Adaptive Single-End Scheme for Detecting and Locating Open Conductor Faults in HV Transmission Lines 6- Abdulla Moselhi (2020) ● EARTHING PROBLEMS IN HIGH VOLTAGE NETWORK SYSTEMS 7- Ali-Algabory (Iraqi): (2019) ● Improving Distance Relays performance in FACTS Compensated TL

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8- Bahaa El Din Hassan E. Soudy (2008)
- A Proposed Technique for Performance Enhancement of Distance Protection Against Power Lines Open-Conductor Conditions Using ANN .
- Bahaa is now the GM of Protection in Egyptian Electrical Authority.

9- Wael Ismael Wahba (2007)
- A Proposed Algorithm for Distribution Power System Protection Using a Combined Wavelet and ANN Technique.
- Wael is now Assoc. Professor at Fauom University.

10- Dalia Farouk Mohamed Allam (2007)
- A New Design of an Intelligent Digital Distance Protective Relay.
- Dalia is now Assistant Professor at Fauom University.

11- Mahmoud Abd Elmonem Salem (2006)
- Proposed Digital Relaying Algorithms for Solving Current Transformer Saturation Problems.
- Mahmoud is now the head of Korimat Power Station.

12- Doaa Khalil Ibrahim (2005)
- Accurate Fault Location in Underground Cables using Advanced Signal Processing Techniques.
- Doaa is now Full Professor at Cairo University.

13- Mohamed Gamal El-Din Hassan (2005)
- Remote Continuous Monitoring For the Ground Grid Resistance.
- Mohamed is now Assistant. Professor at Al-Shorook Engineering Academy.

14- Mohamed A. Al-Naeem (2004)
- A Modified Monitoring and Cathodic Protection Scheme for Controlling the Reinforced Concrete Steal Corrosion.
- Mohamed is now GM of CP Private Company in Saudi Arabia.

15- Omar Hassan (2004)
- Early Detection of HV Insulators Problems.
- Omar is now Assoc Professor at Aswan Energy Inst.

16- Ahmed Abd-El-Moneim Kassem Rizk (2002)
- "A New Digital Current Differential Protective Technique for HV Power Cables."
- Ahmed is now GM in EgyptAir.

M.Sc. thesis advisees (Main Supervisor)

1. Mostafa Mahmoud: (2020): Islanding Detection Using Rate of Change of Zero Sequence of Second Harmonic Voltage
2. Mohamed Yousef: (2020):Detecting and Locating Winding Faults in Power Transformers
3. Mohamed Abd-El-Aziz (2019), " APPLYING Electric Spring Technology in Small Scale Residential Microgrid", (2020)
4. Shady Salah , (2019): SMART DISTRIBUTION SYSTEMS
5. Sameh Ashraf Mansour, (2019) Future of Solar Cooling in Egypt
6. Mahmoud Mehana, (2019): False Trips in Gas Power Plant Due to Transient Voltage Variation.
7. Ahmed Hassan, (2019), Adaptive Multi-Function Transformer Differential Protective Relay.
8. Ahmed Abd-Elmajeed El-Henawy, (2018) : "Harmonic mitigation and Cancellation in Power Distribution Networks"
9. Ahmed Abd-el-Rahman (2017) " EVALUATION OF PROTECTIVE SCHEMES FOR GRID-CONNECTED GENERATOR-TRANSFORMER UNITS IN EGYPT
10. Ahmed Ragab (2017) " STUDYING THE EFFECTS OF DIFFERENT WIND FARM TOPOLOGIES ON SWITCHING OVER VOLTAGES"
11. Eman El-Shaar (2017) " Optimum SELF-Restoration in Egyptian 66 kV Network"
12. Abd-El-Monaem Alsaway (2017) " CENTRALIZED SELF RESTORATION OF ELECTRIC DISTRIBUTION SYSTEMS "
13. Hossam Sabra (2017) " Sympathetic Tripping in Distribution Networks, Causes, Analysis, and Solution"
14. Ahmed Foaad (2017) " Harmonic Resonance Overvoltage during Main Power Transformer Energization in Grid-connected Wind Farms"
15. Mohamed Dawood " Restoring Recloser-Fuse Coordination in Radial Distribution Networks with Distributed Generation "
16. Nehad Khtab (2017) " Development of Zero- Energy -buildings in Egypt "
17. Mohamed Gaber (2016) " Fault Location Identification for Overhead Radial Distribution Networks"
18. Mahmoud Abd-ElFatah (2016) " A Travelling Waves-Based Fault Location Scheme For Multi-Tapped Overhead Distribution Systems"
19. Ahmed Naser (2016) " Fault Zone Discrimination and Fault Location for Series FACTS-Based Transmission Lines"
20. Galal Fathy (2015) " Voltage Stability Enhancement for Radial Distribution Networks".

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21. Emad Francis (2007) "Control and Monitoring for Non-Manned Substation
22. Saber Mohamed Saleh Salim (2005) Performance of Capacitor-voltage transformer using Artificial neural network
23. Walid Ahmed-Fouad " (2003) " A Proposed Zero-Sequence Current Correction for Distance Protection of Double-Circuit Transmission Lines in Meshed Power Systems "
24. Doaa Khalil Ibrahim (2001) "A Digital Communication Based Technique for Distribution High Impedance Fault Detection"
25. Bahaa El Din Hassan E. Soudy (2000)" New Approach for Solving Distance Relaying Setting Problems " .
26. Hany Mahmoud Abdel-Aziz (1999) " Applying Current Limiting Devices on Distribution Systems"
27. Mahmoud Abd Elmonem Salem (1999) " A combined Digital Relay for Power Transformer Protection" .
28. Gehan Abd-El Wahab Mostafa Khalil (1999)" A New Technique for Eliminating Magnetizing Inrush Current Problem in Transformers' Protection" .
29. Samer Ahmed Fouad El-Sokkary "(1999) Digital Multi-Function Relay for Synchronous Generator Protection".

Other Dissertations (External Examiner) (Only from 2014-2022)

(34-thesesis in 9 Universities : Cairo , Alexandria, Ain Shams, Al-Azhar, Helwan, Zagazeg, Tanta, Faoum, Shebeen ElKoom)

1. Amer Elwany (2021) : Power Quality Enhancement for PV Grid-Connected System (Helwan University)
2. Mahmoud ELGamasi (2021): Protection of HVDC Systems for Integrating Renewable Energy Resources (Shebeen EL-Koom)
3. Neven Mahmoud (2020) STATES AND POWER CONSUMPTION ESTIMATION FOR NILM. Cairo University, Communication Dept.
4. Dalia Arafa (2020), NOVEL OPTIMIZATION TECHNIQUES FOR A PROPER SOLAR SYSTEM DESIGN , Ph D, Fayoum University
5. Nora Ahmed (2019) LRA for Short term Load Forecasting, Zagazig University
6. Naheel Shaker, (2019) An Intelligent Protection Scheme for active distribution networks. Alexandria University
7. Eatemad Wageh, (2019) New Protection Scheme for Low Voltage DC microgrids, Tanta University.
8. Protective relay coordination with DG, (2018) M.Sc. Alexandria University
9. Arc Furnace , (2017) Ph D. , Alexandria university.
10. OPTIMIUM SITE SELECTION OF Large SCALE SOLAR POWER PLANTS, Janet Aid (2017) M.Sc. Helwan University.

11. Three Winding Power Transformer Protection Based On Pattern Recognition Techniques, Abdullah Mahomed (2016) M.Sc. Helwan University.
12. Induction Generator Protection Using Wavelet Analyses Technique, Ahmed Tamam (2016) M.Sc. Helwan University.
13. PV Parameters Estimations, Dalia Mahmoud (2016) M.Sc. Fayoum University.
14. Protection Of Electrical And Electronic Equipment And Electronic Parts Against Electrostatic Discharges, Amany Almasry (2016) M.Sc. Al-Azhar University.
15. Proposed Schemes for Enhancing Coordination Of Directional Overcurrent Relays In Meshed Distribution Systems With Dg Units - Phd., Heba Sharaf (2015) ,Cairo University.
16. Distance Relay For Load Encroachment Alleviation With Agent Based Supervision Of Zone-3, Mohamed Badr, M.Sc. (2015) , Alexandria University
17. Simulation And Analysis Of Smart Micro Grid, Mohamed Adel (2015) M.Sc. Helwan University
18. Identify The Fault Location In Electrical Distribution Network By Calculation The Impedance Value By Using Neural Networks, Ali Husin Alkalili (2015) , M.Sc. Zagazek University
19. Demand And Generated Power Balancing Of Power Systems With Large Scale Renewable Energy Sources , Ahmed Rahoma (2015) , M.Sc. Helwan University
20. Enhancement Of Grid-Connected Wind/Pv Hybrid System Using STATCOM, Mohamed Bahi-Barakat , M.Sc. Cairo University
21. Wind Energy Water Pumping System With Storage, Abdelmonem Shaaban (2015) , M.Sc. Helwan University
22. Optimal Design Of Protection Schemes For Distribution Networks Including Distributed Generation, Ahmed Adel Abozied (2015) , Ain Shams University
23. Effect Of Harmonics On Electromechanical And Solide-State Relays Performance, Mohamed Saleh (2015) , M.Sc. Fayoum University.
24. Intelligent Load Management System Using Monitoring And Control System For Industrial Facilities, Yahya Mahmoud (2014) , M.Sc. Cairo University.
25. A New Directional Relay Based On Active / Reactive Power Portrait , Shreef Said (2014), M.Sc. Helwan University
26. Enhancing Distance Relay Performance For Transmission Line Protection, Mohamed Essam (2014) , M.Sc. Cairo University
27. Comprehensive Fault Location Scheme For Power Transmission Lines", Ahmed Galal (2014) , M.Sc. Cairo University.
28. Analysis And Elimination Of Capacitors Switching Transient Overvoltages In Power Factor Improvement Of Induction Motors, Mohamed Ossama,(2014) , M.Sc. Cairo University.

	29. HVDC Transmission, Marwa Attea (2014) , M.Sc. Helwan University 30. An Approach For Achieving Energy Credits In An Egyptian Green Building Rating System, Ahmed Ibrahim (2014) , M.Sc. Cairo University 31. Islanding Phenomenon Detection Of Distributed Generation, Ahmed Gamal (2014) , Fayoum University 32. Design Of A Prototype Cable Fault Location, Moamen Mohamed (2014) , M.Sc. Fayoum University 33. Improving Voltage Stability Of Wind Farms Connected To Weak Grids Using Facts, Mohsen Galal (2014), M.Sc. Cairo University 34. Energy Credit Investigation In The Development Of An Egyptian Green Building Rating System, Ahmed Ali (2013), Cairo University
Funded Research Projects.	○ Cost Reduction of Solar Energy Projects (Cairo University) 2002 ○ New Methods for fault detection in Kuwait Electrical Networks. (The Public Authority for Applied Education and Training) 2006 ○ Improving the Performance of Backup Protective Relays in Kuwait Transmission Networks. (Kuwait Foundation for the Advancement of Sciences) 2007 ○ New Techniques for Estimating the Distance of Open-Circuit Faults Using Distance Relays (Kuwait Foundation for the Advancement of Sciences) 2008 ○ Short Circuit Level Reduction in the Kuwait Developed Distribution Networks (The Public Authority for Applied Education and Training) 2009 ○ A Wavelet Based Technique For Detection and Classification of Transients in Distribution Networks . (The Public Authority for Applied Education and Training) 2010
Official Committees	○ **Member in the Following Official Technical Committees for:** ○ Studying the effect of current interruption and voltage reduction on the industrial plants in Egypt. (Scientific Research Academy, 1994) ○ Evaluation of the value of Transferring the Technology of Dry Transformers to Egypt . (Capital Stock Authority) 1998 ○ Examining the performance of a new digital relay (AUOS) before being used by the Egyptian Electrical Authority in the Egyptian distribution networks. The work include conducting all types of protective relay testing, field testing at different transmission and distribution lines and examining the relay SCADA system.(Ministry of Electricity, Egypt) 2002

	○ Preparing a technical report to an Egyptian Court on the Conflict between the Ministry of Treasure and Power-Egypt Company. (Ministry of Justice) 2001
International Activities	○ Named In Who's Who. ○ Keynote Speaker in First International Power and Energy Conference (Malaysia 28-29 Nov. 2006). ○ Member of the Technical Committee of First International Power and Energy Conference (Malaysia 28-29 Nov. 2006) ○ Winner of the best paper Award in GCC – Cigre 2004. ○ Member of the Technical Committee of IASTED Conference on Energy and Power Systems from 2003 till 2009. ○ Member of the Technical Committee of Power Simulation Conference, Amest 12-14 Oct, 2003.
International Conference	○ DPSP-2018, Belfast, 12-15 March, 2018. ○ MEPCON-2016, Cairo, 17-19 Dec. 2016 ○ ICEES-2010, Singapore, Feb. 26-28, 2010. ○ WASET-2009, Tokyo, Japan, May 27-29, 2009. ○ CIRED 2008: Smart Grids for Distribution, Frankfurt, 23 - 24 June 2008. ○ 7th IASTED International Conference on Power and Energy Systems (EuroPES), Aug 29-31, Palma De Mallorca, Spain, 2007. **Chairman of the Transmission/ Protection Session** ○ 15th International Conference on Power System Protection, Bled, Solvenia, Sept. 6-8, 2006. ○ First International Power and Energy Conference PECon 2006, November 28-29, 2006, Putrajaya, Malaysia. **Chairman of the Transmission/ Protection Session** ○ 14th International Conference on Power System Protection, Bled, Solvenia, Sept. 29-October 1, 2004. ○ GCC CIGRE Regional Conference on Transmission Line Design and Operation, Kuwait, 1-2 Dec. 2004. (**Best Paper Award**) ○ The 20th IASTED Conference on Power and Energy Systems (Euro Power Engineering Society -2002), Crete, Greece, 25-28 June, 2002. ○ MEPCON-98, Mansora, 15-17 Dec. 1998 ○ MEPCON-97, Alex, 4-6 Jan. 1997 ○ ICECS'97, Cairo, 15-18 Dec. 1997 ○ Cigre-97, Session, Cairo ○ AEIC.95, Cairo, 16-19 Dec. 1995
Reviewer in International Journals	○ IEEE Transaction on Dielectrics and Electric Insulation ○ IEEE Transaction on Power Delivery ○ Journal of Electrical Power and Energy Systems, ○ International Journal on Emerging of Electric Power System. ○ Journal of Electrical Power Components. ○ European Transactions on Electrical Power

	○ Ain Shams Engineering Journal ○ International Journal of Emerging Electric Power Systems.
Scholastic Awards	○ Prize of Best Practical Ph. D. Supervised Thesis from the College of High Studies, Egypt (2008) ○ Canadian International Development Agency (CIDA) Award for a 3 Years Ph.D. Study at The University of Calgary, Canada (1990) ○ Academic Excellence Scholarship, Cairo University 1988) ○ Cairo University Graduate With Honor Degree Award (1987) ○ Scholarship of Distinction, Cairo University, 1982-1987. ○ Cairo Provincial Award for Egypt's 10-Top Students. (1979)
Published Books	○ **In English** ○ Chapter, Leading-Edge Electric Power Research, ISBN 978-1-60021-981-8, © 2007 Nova Science Publishers, Inc. ○ **In Arabic**)Free Download from www.drgilany.com **with more than 300000 downloads**) ○ The hard copy for the 5-Books are published in 2020 (Third Edition), دار المراجع العلمية. ○ Power System Protection (2007) ○ Electrical Installations (2010) ○ Power Transformers (2013) ○ Electrical Power Engineering (2016) . ○ Light Current Systems (2020)
Published Papers:	635 Citation on research gate (4191 Reads) : https://www.researchgate.net/profile/Mahmoud_Gilany h- index 19 on google scholar- Citations 840 https://scholar.google.com/citations?view_op=list_works&hl=en&user=xLsd46MAAAAJ h-index 13 (566 Citations) on Scopus Preview https://www.scopus.com/authid/detail.uri?authorId=24448081500 **URL :**http://staff.eng.cu.edu.eg/drgilany https://orcid.org/0000-0002-4870-4352 **93 Published Papers in Journals and International Conferences.**

1- Mohamed Esam El-Dine Atta; Doaa Khalil Ibrahim; Mahmoud Gilany;' Broken Bar Fault Detection and Diagnosis Techniques for Induction Motors and Drives: State of the Art', IEEE access , Volume 10, Aug 2022, pp 88504 - 88526. ,
DOI: 10.1109/ACCESS.2022.3200058, 2022.

2- Mohamed Esam El-Dine Atta; Doaa Khalil Ibrahim; Mahmoud Gilany; Ahmed F. Zobaa,' Adaptive Scheme for Detecting Induction Motor Incipient Broken Bar Faults at Various Load and Inertia Conditions', MDPI Sensors, Volume 22, Issue 1, 365, 2022. https::doi.org/10.3390/s22010365.

3- Loai Mohamed, Doaa Khalil, Mahmoud Gilany," Enhancing Distance Relay Performance Using Wide-Area Protection for Detecting Symmetrical/Unsymmetrical Faults during Power Swings", Alexandria Engineering Journal (2022) 61, 6869–6886.
https://doi.org/10.1016/j.aej.2021.12.031

4- Mohamed Esam El-Dine Atta, Doaa Khalil, Mahmoud Gilany," Detection and Diagnosis of Bearing Faults under Fixed and Time-Varying Speed Conditions Using Persistence Spectrum and Multi-Scale Structural Similarity Index", IEEE Sensors Journal ,page(s): 2637-2646, ISSN: 1558-1748, Vol 22, Issue 3, Feb., 2022.
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